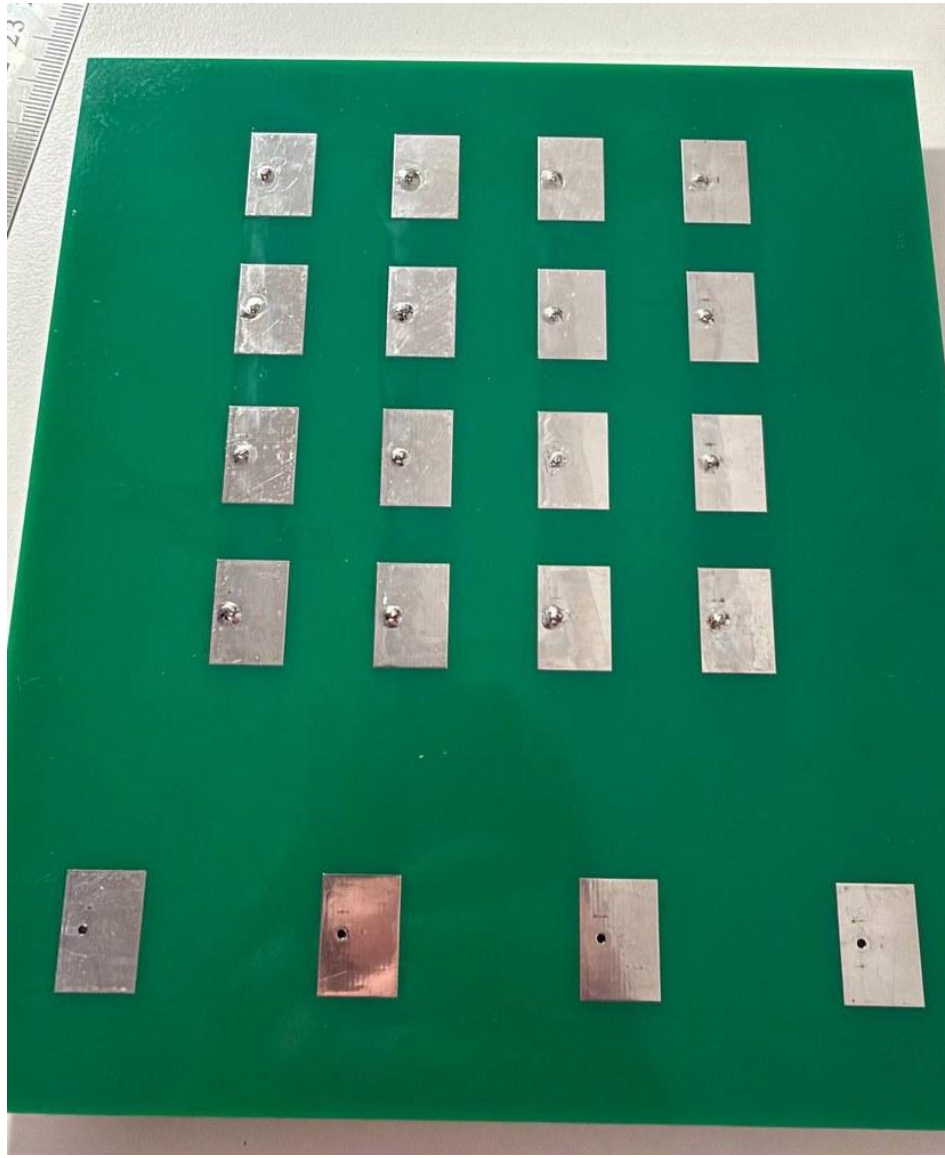
A 4×4 phased-array research platform built from ESP32-C5 radios

5.2 GHz · two 8-element tiles · PARLIO aggregation into one ESP32-S31 per tile · 1 GbE per tile

Student project proposal — technical concept

October 2026

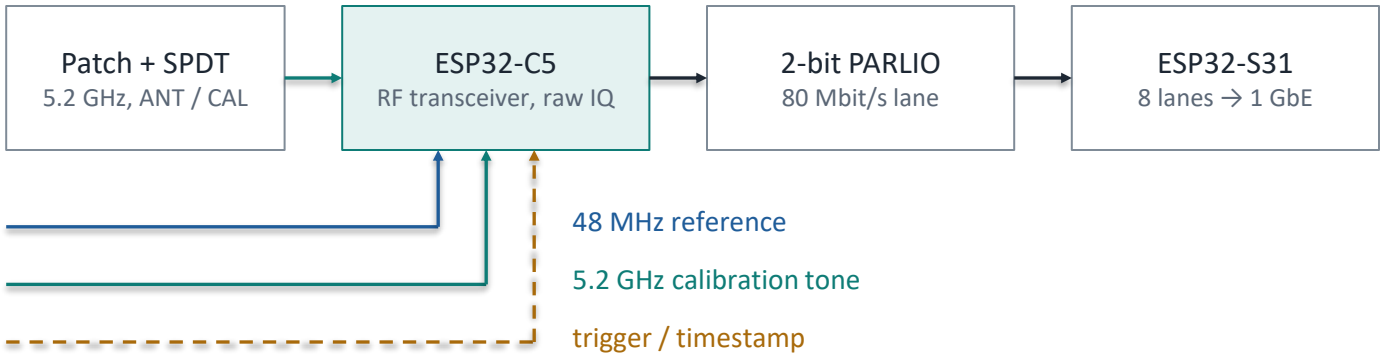


Builds on the open ESPARGOS and ESP-SDR projects. Photo: our existing 4×4 patch board (5.2 GHz, FR4).

What are we trying to build?

A 16-element digital array at 5.2 GHz in which every element is a complete ESP32-C5 transceiver. The aim is a research platform for receiving, sensing and communication, with coherent transmission as a later research question.

One element (×16)





4 × 4 elements
λ/2 pitch at 5.2 GHz
no grating lobes for ±90° scan

Shared by all 16 elements — this is what turns 16 radios into one array.

● Receive

digital beamforming, AoA / direction finding (shown by ESPARGOS at 2.4 GHz)

○ Sense

Wi-Fi sensing, passive and bistatic radar, Doppler

○ Sound

channel sounding, MIMO measurements, localization

◌ Transmit coherently

TX beamforming, if raw TX can be reproduced

● Confirmed ○ Engineering inference ◌ Speculative

λ = c / 5.2 GHz = 57.7 mm; d = λ/2 = 28.8 mm. Status = shown on ESP32-class hardware (ESPARGOS, 2.4 GHz) — none of it is shown on the C5 at 5 GHz yet.

What already exists, and what is new here

ESPARGOS (Univ. of Stuttgart)

- ESPARGOS One: 8 ESP32-C61 sensors + a controller ESP32; 4 × 2 dual-polarized patches, 2.4 GHz
- shared 40 MHz clock + 2.4 GHz reference through a splitter and an RF switch at every sensor
- PLL phase changes after every reset and channel change, so calibration is repeated per configuration
- raw IQ (closed firmware): wired trigger, packet timestamps ≤ 37.5 ns apart, CW tone from the controller's PLL
- up to 4 boards (32 elements) combined; clock + reference share one SMA cable in a star

ESP-SDR

- raw IQ from an undocumented modem sample-dump engine; 10 chips; open source since Sept 2026
- ESP32-C5 receives at 5 GHz: a 5.5 GHz test tone, and a second C5 on channels 36/48 (5.17–5.25 GHz)
- C5: snapshots of ≤ 16,380 samples at up to 80 MS/s, 10-bit I/Q, sent over USB or UART
- software trigger only; no timestamps, no multi-device synchronization
- receive-only; ESP-SDR's CPU spectrum mode on the C5 returns power, not complex bins

What this project adds

- the ESPARGOS synchronization principle at 5.2 GHz with a 48 MHz reference
- one external 48 MHz clock driving 16 chip-down ESP32-C5s; C5 LO-phase behaviour
- hardware trigger and capture alignment for C5 snapshots
- IQ export over 2-bit PARLIO into one ESP32-S31 per tile, then 1 GbE
- in-band 5.2 GHz calibration network; two 8-element tiles with a master timing board

Confirmed · Engineering inference · Open — needs bench test · Speculative

Sources: ESPARGOS One datasheet (June 2026), espargos.net, pyespargos, arXiv 2502.09405; ESP-SDR page and repository (HEAD 2026-10-08).

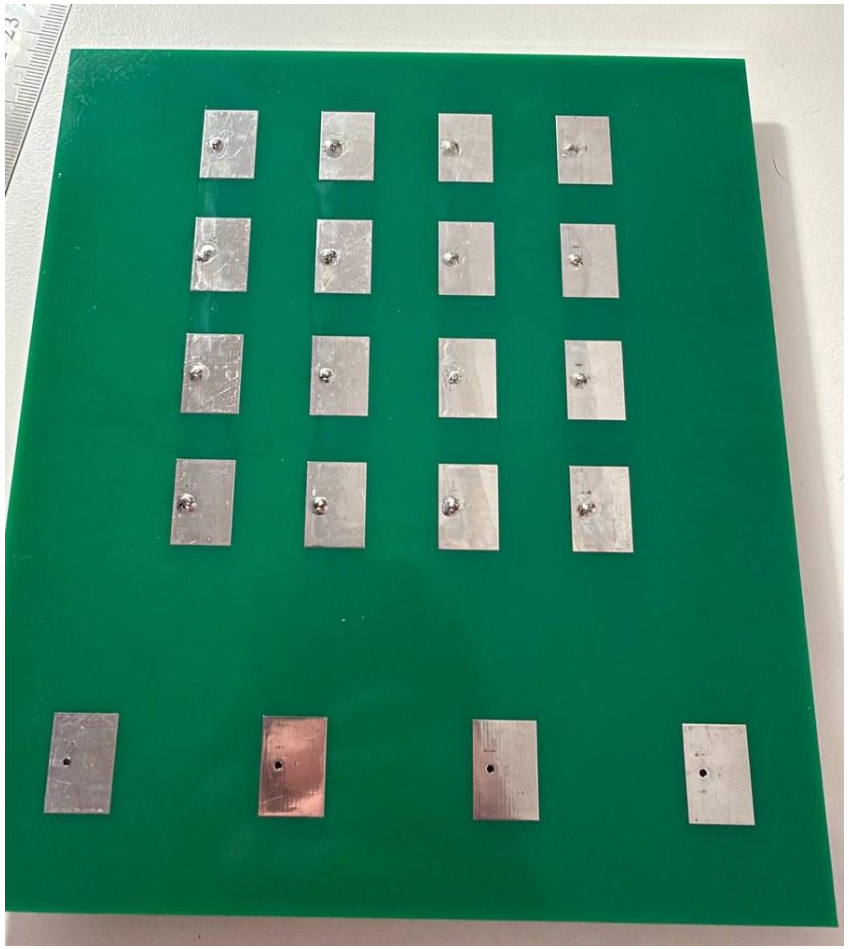
Why ESP32-C5?

Chip	Wi-Fi	5 GHz	Main crystal	Relevance here
ESP32-C5	Wi-Fi 6, 2412–2484 + 5180–5885 MHz	yes	48 MHz	only 5 GHz part; separate ANT_5G pin; ESP-SDR snapshot IQ
ESP32	Wi-Fi 4, 2.4 GHz	no	40 MHz	2.4 GHz only
ESP32-S2 / S3	Wi-Fi 4, 2.4 GHz	no	40 MHz	2.4 GHz only
ESP32-C2 / C3	Wi-Fi 4, 2.4 GHz	no	26 or 40 / 40 MHz	2.4 GHz only
ESP32-C6 / C61	Wi-Fi 6, 2.4 GHz	no	40 MHz	C61 is the sensor in ESPARGOS One
ESP32-H2	none (BLE, 802.15.4)	no	32 MHz	no Wi-Fi; ESP-SDR still receives
ESP32-S31	Wi-Fi 6, 2.4 GHz	no	40 MHz	continuous IQ streaming in ESP-SDR (experimental), 2.4 GHz only

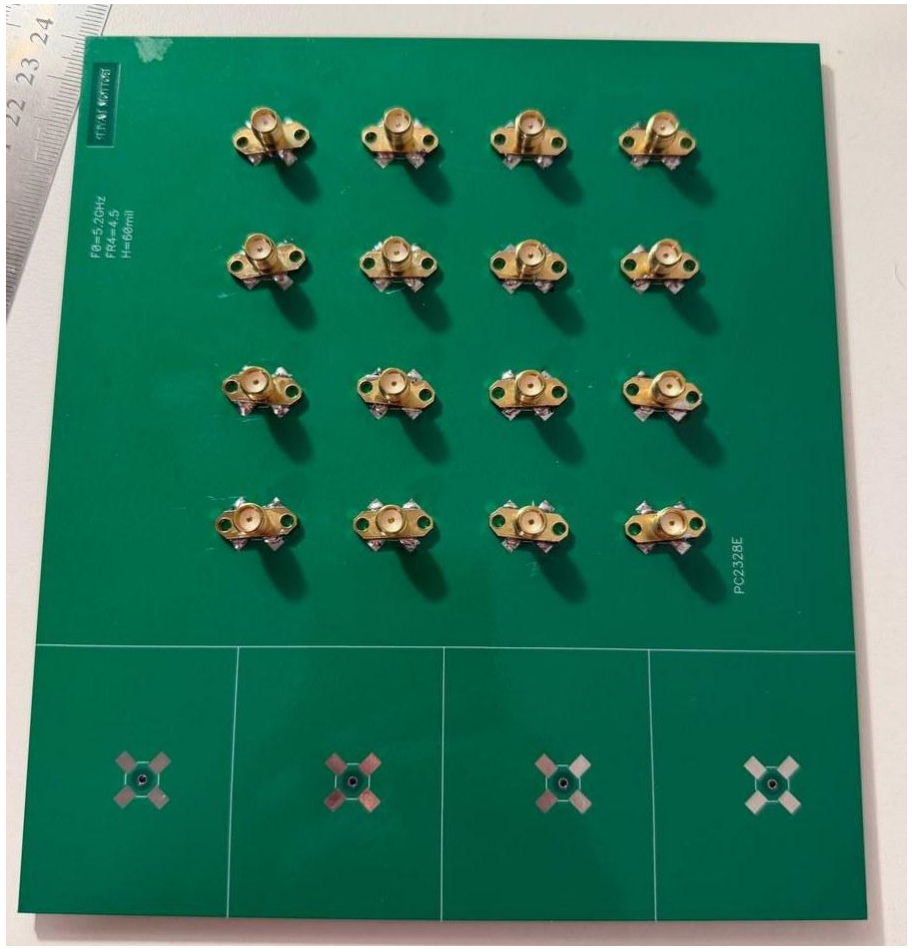
Only the ESP32-C5 reaches 5.2 GHz natively. ESP-SDR receives on all ten parts, but every other chip would need an external mixer and LO per element to reach 5.2 GHz — which defeats the point of a low-cost array. The price: the C5 has no continuous IQ path today.

Sources: Espressif datasheets (C5 v1.5, S31 v0.5 pre-release, others current); ESP-SDR README chip-support table. Checked 12 Oct 2026.

The antenna board we already have



radiating side: 16 probe-fed patches + 4 test patches



back: one SMA per element

What we can read off it

- 4 × 4 rectangular patches, linear polarization
- silkscreen: f_o = 5.2 GHz, FR4 εr = 4.5, h = 60 mil
- pitch ≈ 30 mm from the ruler, consistent with λ/2 = 28.8 mm
- expected patch ≈ 13 × 17 mm — matches the proportions

What it means

- Rev 1 can use this board as the aperture and cable each element to its tile — no seam crosses the radiating surface.
- Or the tiles carry their own patches (next section) — then substrate thickness and the seam need care.
- Either way, measure S11 and neighbour coupling per element first.

Observations from photos; pitch and patch size to be confirmed against the layout files. Patch estimate: transmission-line model, εr = 4.5, h = 1.52 mm.

The hard part is synchronization

Sixteen independent radios disagree in frequency, phase, timing and gain. A shared clock fixes only the first of these; the rest have to be measured and removed.

Error term	Where it comes from	Shared 48 MHz clock	5.2 GHz cal. tone	Common trigger
Carrier frequency offset, drift	independent crystals (tens of ppm)	removes	—	—
LO phase offset	each PLL locks with an arbitrary phase	—	removes	—
Sampling-time offset	independent sample clocks and start times	partly	removes (swept tone)	partly
Capture start misalignment	firmware / interrupt latency	—	partly	removes
Fixed path mismatch	cables, PCB traces, switches, filters	—	after the switch	—
Gain mismatch	AGC state, LNA spread	—	if gain is fixed	—
Uncorrelated phase noise	additive jitter of buffers, PLL noise	partly	—	—

Consequence: calibration is not a one-off factory step. The LO phase changes every time a PLL relocks, so the array must be able to re-measure itself on demand — that is why the calibration network is part of the hardware.

Error model follows the ESPARGOS approach (shared clock + in-band phase reference + digital correction); C5-specific behaviour must be measured.

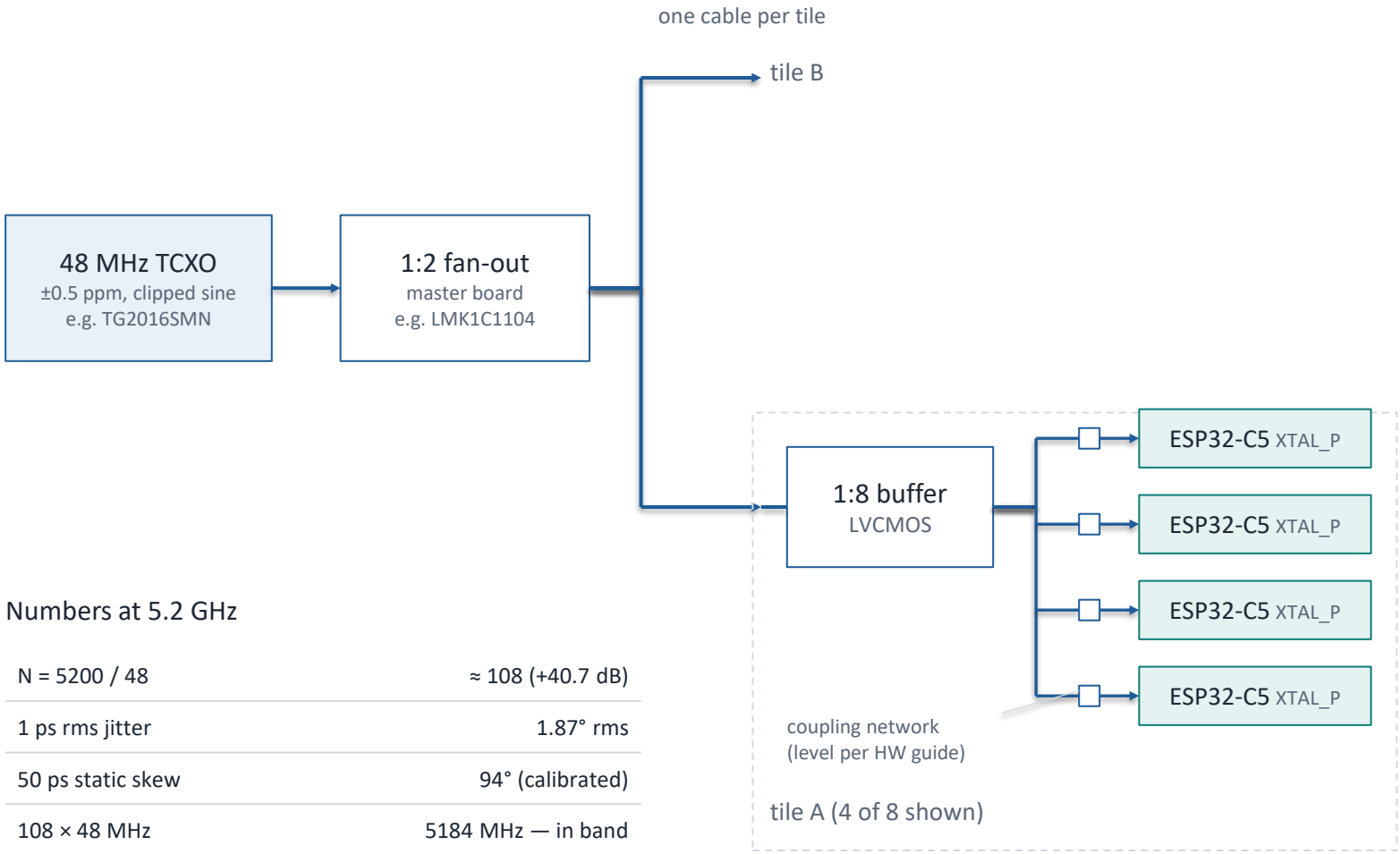
Four synchronization domains — and what Ethernet is not for

Domain	Signal and path	Guarantees	Does not guarantee
RF reference	48 MHz TCXO → fan-out → 16 × XTAL_P	one frequency for all LOs	LO phase after each lock
Data transport	40 MHz PCLK → fan-out → 16 C5 + 2 S31	lane-to-lane cycle alignment	which sample a lane starts with
Capture trigger	master GO → S31 → PCLK enable (+ READY)	common start within a tile	tile-to-tile start (two S31s)
RF calibration	5.2 GHz tone → 1:2 → 1:8 → SPDT	measures ϕ_k, τ_k and path mismatch	anything before the switch
Ethernet / PTP	S31 EMAC, 1588 timestamps	logging, tile coordination	sample or phase alignment

Wideband correction per element is a phase plus a delay: $C_k(f) = \exp[-j(\phi_k + 2\pi f\tau_k)]$, applied per FFT bin before the channels are combined.

Team architecture; ESPARGOS uses the same split (shared clock, wired trigger, switched RF reference). Ethernet / PTP is for coordination only.

Sharing one 48 MHz reference



Numbers at 5.2 GHz

N = 5200 / 48	≈ 108 (+40.7 dB)
1 ps rms jitter	1.87° rms
50 ps static skew	94° (calibrated)
108 × 48 MHz	5184 MHz — in band

Frequency

All 16 LOs derive from one oscillator, so carrier offsets between elements vanish. The absolute ppm error is common to all and harmless.

Static skew

Buffer skew becomes a fixed LO phase: 50 ps ≈ 94° at 5.2 GHz. Calibration removes it — but skew drifting with temperature shows up as phase drift.

Jitter

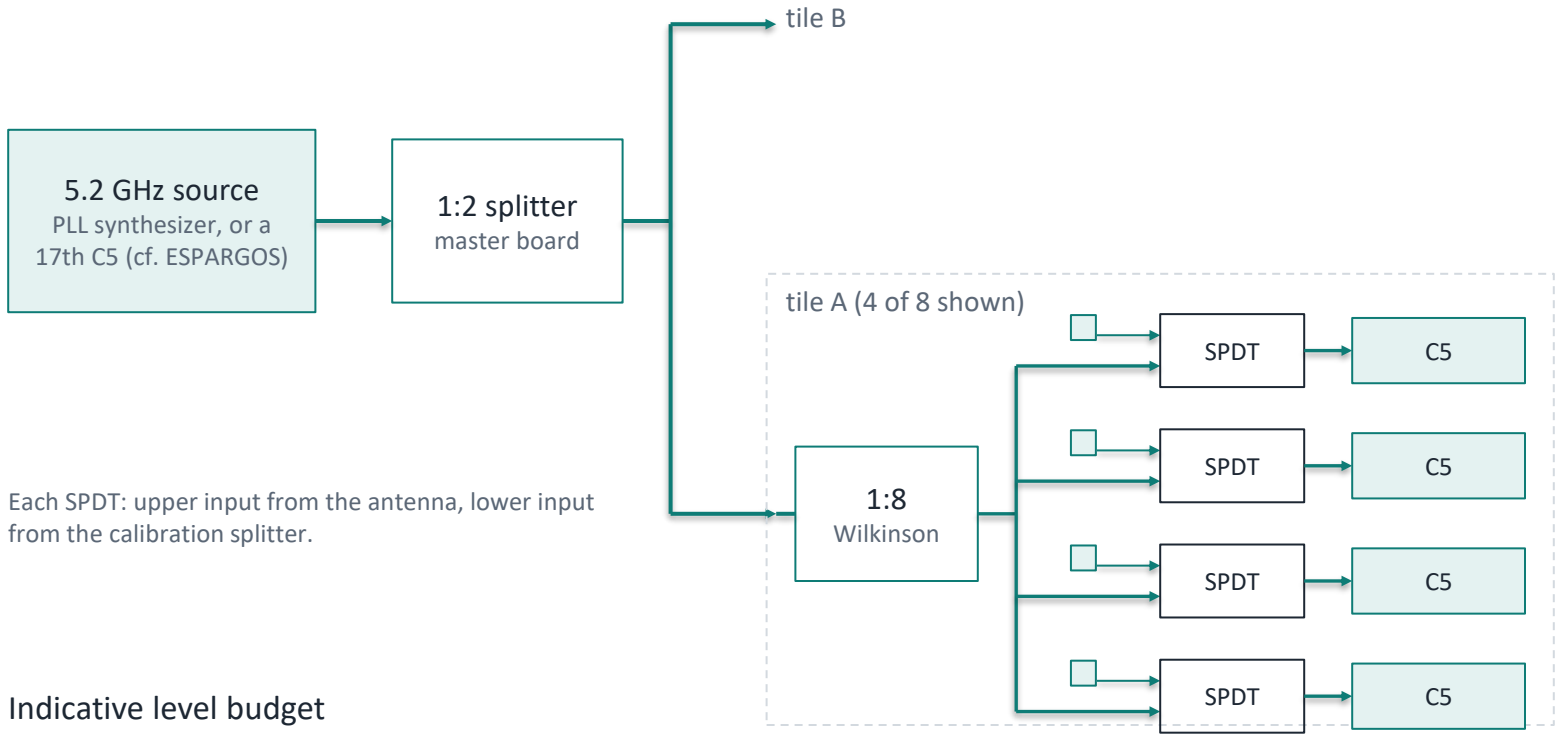
In-band reference noise rises by 20·log(5200/48) = 40.7 dB. Jitter common to all elements cancels in relative phase; buffer additive jitter (20–160 fs max ≈ 0.04–0.3°) does not.

Injection

The C5 design guidelines cover only a 48 MHz ±10 ppm crystal, with no external-oscillator requirements (the datasheet calls XTAL_P/N a crystal or oscillator input). Precedents: the ESP32-S2 guide (oscillator into XTAL_P via ~50 pF), ESPARGOS (40 MHz shared), Espressif's esp-crab (two C5s, "co-crystal"). C5 modules carry their own crystal.

ESP32-C5 HW design guidelines (48 MHz only, ±10 ppm, >500 mV, 24 nH series L); ESP32-S2 HDG oscillator note; TI SWRA029 (20·log N); buffer and TCXO datasheets.

A 5.2 GHz calibration network



Each SPDT: upper input from the antenna, lower input from the calibration splitter.

Indicative level budget

Stage	Loss
1:2 + 1:8 split (ideal 3 + 9 dB)	≈ 14–15 dB
cables + ≈ 18 cm of tile lines	≈ 5–7 dB
SPDT insertion loss	≈ 1 dB
source 0 dBm → at each C5	≈ −20 to −23 dBm

Well below the C5's +5 dBm maximum input, but enough to clip the 10-bit samples at high gain: attenuate and use one low, fixed gain index. SKY13351-class switches specify ≈ 24 dB isolation (at 2.45 GHz; likely less at 5 GHz), so mute the synthesizer during antenna captures.

Global source or one per tile?

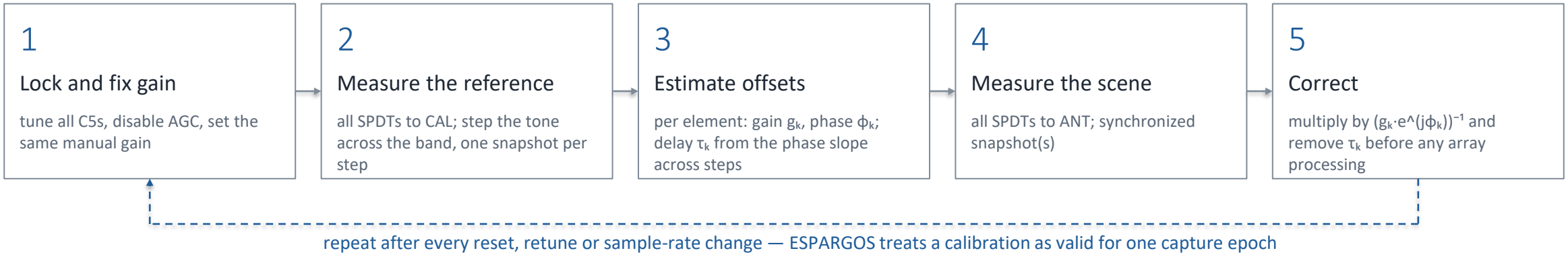
	Global	Per tile
Inter-tile phase reference	yes, common	none — needs OTA step
RF routing	2 cables	local only
Cost / parts	1 source	2 sources
Tiles testable alone	with a jumper	yes

Proposal: one MAX2871-class source on the master board, a 1:2 split to the tiles, a 1:8 Wilkinson tree and eight SPDTs per tile. Each tile also gets a local test input so it can be calibrated alone during bring-up.

○ architecture choice, adapted from ESPARGOS

Adapted from ESPARGOS. MAX2871: VCO 3–6 GHz, −1 to +8 dBm, mutable outputs. SKY13351: 0.35 dB loss, 24 dB isolation at 2.45 GHz (datasheet); less expected at 5 GHz.

Calibration is a measurement loop



Per-element model during a snapshot

$$y_k[n] = g_k \cdot e^{j\varphi_k} \cdot x(nT_s - \tau_k) + w_k[n]$$

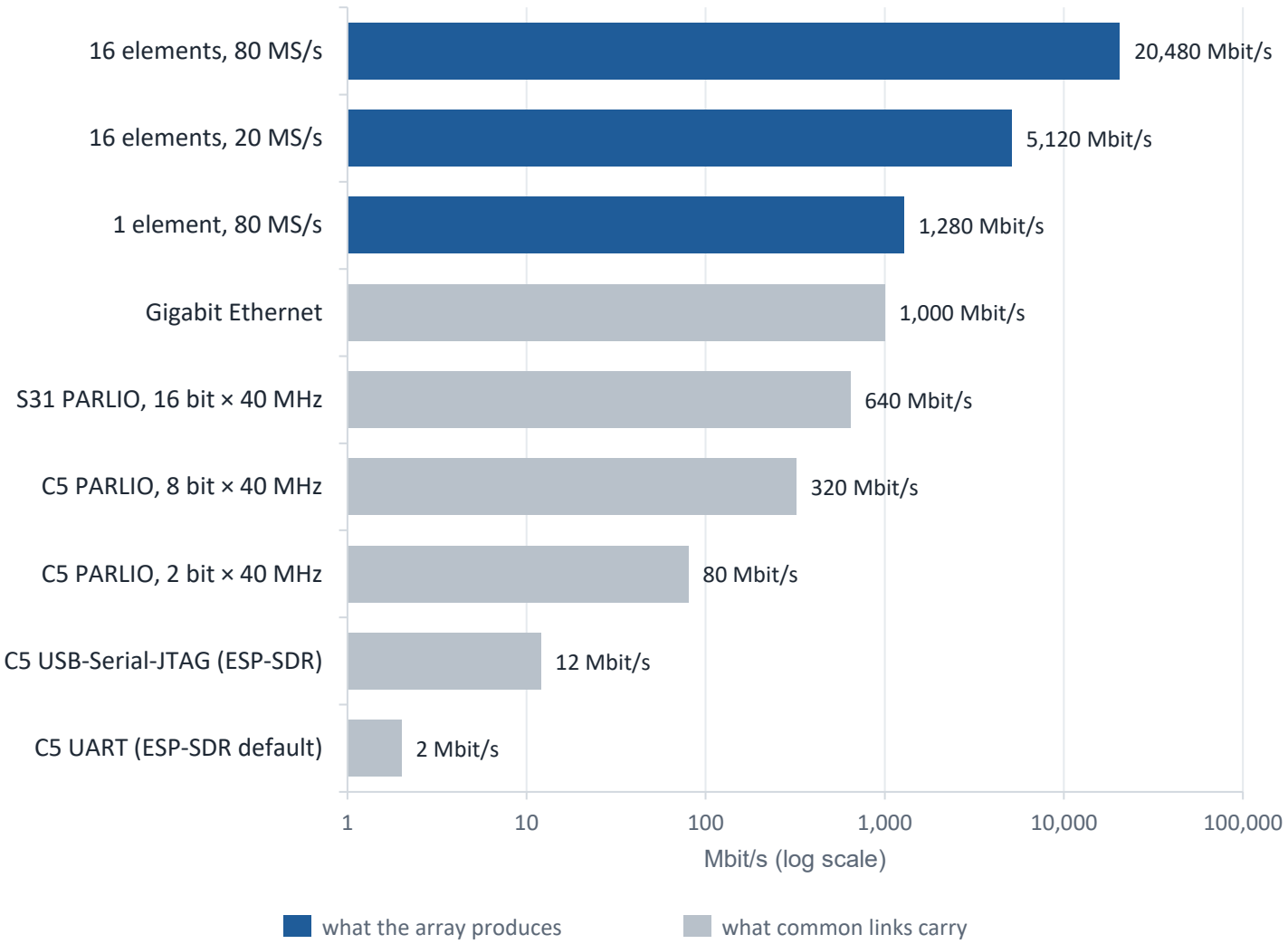
In CAL mode x is the common reference, so g_k, ϕ_k, τ_k follow from comparing each y_k with element 1. A single CW tone cannot separate ϕ_k from τ_k; ESPARGOS sweeps the tone across the band and fits both, after coarse alignment on a shared reference packet.

What the tone cannot see

- Everything before the switch: patch, probe via and a 1 mm line, the same for all eight (or the cable, with a separate aperture). Calibrate once over the air; tile B adds a fixed 180°.
- Mutual coupling between elements: measure embedded element patterns once the first tile works.
- Errors that change faster than the calibration interval.

Procedure follows ESPARGOS One IQ mode (pyespargos iq_sync / iq_pool): packet-timestamp coarse sync, swept CW tone for (τ, ϕ), valid for one capture epoch.

Sixteen receivers create a data problem



16 × 80 MS/s × 16 bit

= 20.48 Gbit/s

The C5 samples at up to 80 MS/s internally, but ESP-SDR moves captures out over USB (12 Mbit/s) or UART.

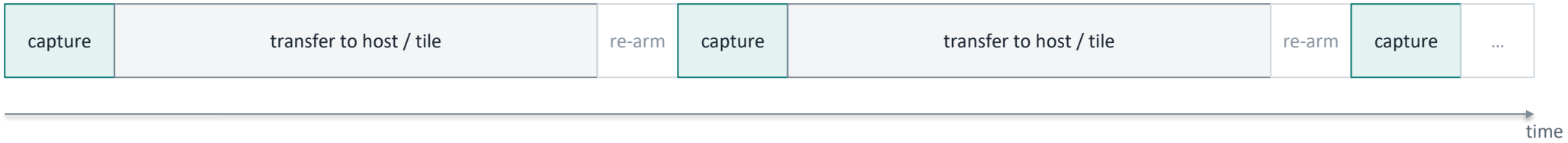
A 2-bit PARLIO lane at 40 MHz gives each C5 80 Mbit/s: enough for ~4 MS/s continuous, or for draining full-rate snapshots in a few milliseconds.

Eight lanes fill one 16-bit S31 bus (640 Mbit/s) and one Gigabit port per tile — a rate ESP-SDR already pushes through an S31 (≈ 687 Mbit/s on the wire). Continuous raw 80 MS/s from 16 elements stays out of scope.

Line rates from datasheets; ESP-SDR uses UART / USB on the C5 today. PARLIO figures are width × 40 MHz. Raw IQ as 8-bit I + 8-bit Q.

What ESP-SDR gives us today: snapshots, not streams

One element, snapshot mode



The largest C5 snapshot is 16,380 complex samples: 204.75 µs at 80 MS/s, 32,760 bytes as 8-bit I/Q. Moving it out takes 100–1000× longer than capturing it, so the duty cycle is set by the output port, not by the ADC.

Best case for one 16,380-sample IQ8 snapshot

Output port (per element)	Transfer time	Snapshots / s	Duty cycle
UART, 2 Mbaud (ESP-SDR today)	163.8 ms	≤ 6	≤ 0.12 %
USB-Serial-JTAG, 12 Mbit/s line	21.8 ms	≤ 45	≤ 0.93 %
PARLIO lane, 2 bit × 40 MHz (planned)	3.3 ms	≤ 287	≤ 5.9 %
PARLIO, 8 bit × 40 MHz (ceiling)	0.8 ms	≤ 977	≤ 20 %

What this means

- Raw snapshots suit calibration research, beamforming and offline algorithm work — the first milestones.
- Doppler and tracking need many snapshots with no retune in between (a C5 retune takes ~28–35 ms at 5 GHz).
- The planned 2-bit PARLIO lane drains a full snapshot in 3.3 ms — about 7× faster than USB at its line rate.

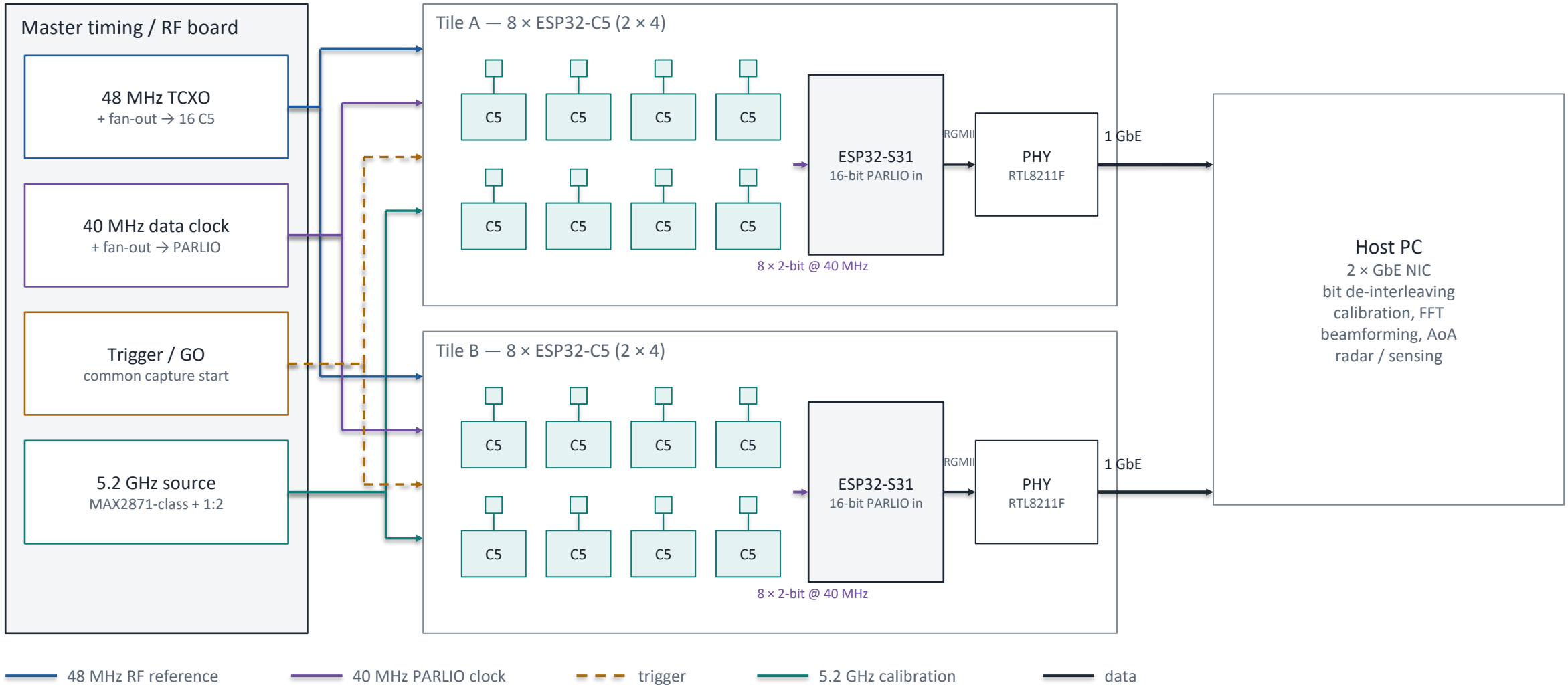
ESP-SDR C5 limits from chip.h (IQ_WORDS = 16,380), README and PR #9 (retune time). Transfer times are upper bounds from line rates — no measured C5 snapshot rate is published.

Five ways to collect the data

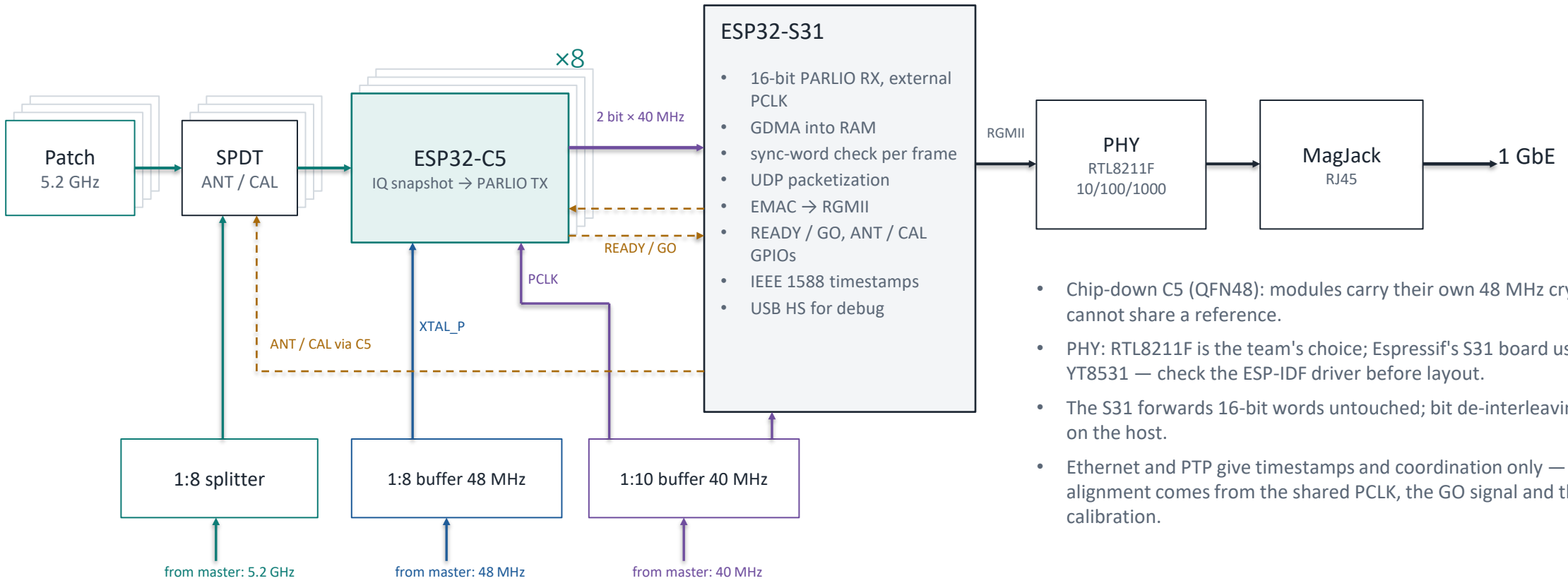
	A · Direct to host	B · Ethernet per C5	C · Central FPGA	D · 4 × 2×2 FPGA tiles	E · 2 × 8 tiles, PARLIO → S31
Topology	each C5 → USB hub → PC	C5 → W5500 → switch → GbE	16 C5 → one FPGA → Ethernet	4 × (4 C5 → FPGA) → hub	8 × 2-bit PARLIO → S31 → RTL8211F → GbE
Good for	bring-up, 1–4 channels	separate networked nodes	deterministic capture, DSP	local reduction, scaling	cheap integrated tile, two uplinks
Data limit	USB full speed per C5	SPI bridge ≤ 100 Mbit per C5	16 C5 slave ports	¼ of the array per tile	80 Mbit/s per lane, 640 per tile
Sample alignment	software only	none from Ethernet	ns, in fabric	ns inside a tile	shared PCLK + GO; to be proven
Hardware effort	low	high: 8 bridges + PHYs + switch	high: dense board	high: FPGA per tile	medium: one S31 + one PHY per tile
Where it fits	step 1	rejected	Plan B variant	Plan B variant	Rev 1 baseline

Effort and suitability are judgements for a student team. E is the current team proposal.

Two 8-element tiles and one timing board



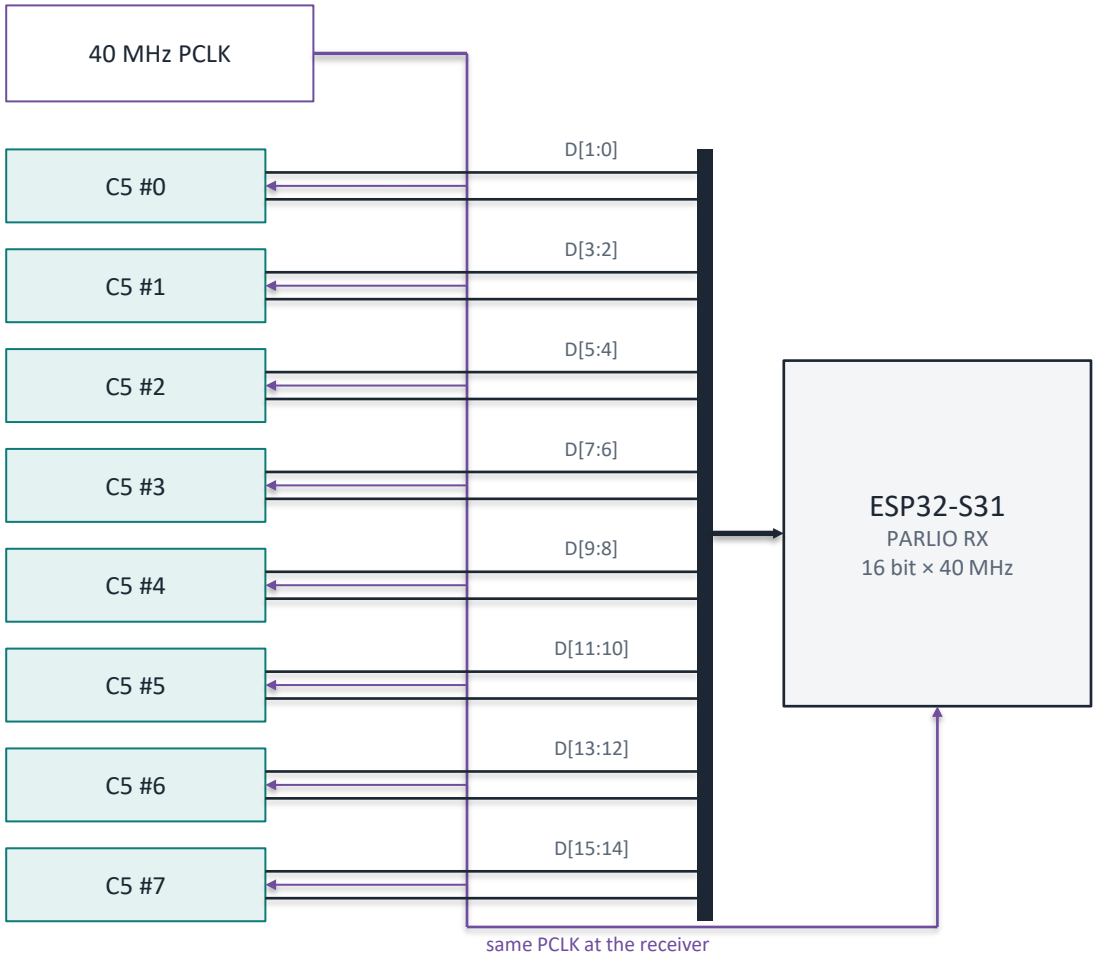
One tile: eight C5s, one S31, one Gigabit port



- Chip-down C5 (QFN48): modules carry their own 48 MHz crystal and cannot share a reference.
- PHY: RTL8211F is the team's choice; Espressif's S31 board uses a YT8531 — check the ESP-IDF driver before layout.
- The S31 forwards 16-bit words untouched; bit de-interleaving runs on the host.
- Ethernet and PTP give timestamps and coordination only — sample alignment comes from the shared PCLK, the GO signal and the RF calibration.

8 × 80 Mbit/s = 640 Mbit/s on one 16-bit bus at 40 MHz — 64 % of a Gigabit link. No Ethernet MAC, PHY, magnetics or switch per element.

Eight 2-bit streams on one 16-bit bus



Start sequence

1. the S31 arms all C5s; each captures a snapshot and loads its PARLIO DMA descriptors
2. each C5 raises READY
3. the S31 waits until all eight READY lines are high
4. GO: the S31 enables the PCLK buffer, so all lanes start on one edge
5. all C5s shift out on the same clock edges
6. the S31 captures 16-bit words by DMA
7. every frame starts with a sync word common to all lanes

Per-lane frame (every C5 sends the same layout)

Field	bits	Purpose
sync word	32	identical on all lanes → alignment check
channel ID	8	which C5 / antenna
frame sequence	16	detects lost frames
sample counter	32	ties the frame to the capture start
capture metadata	32	rate, gain index, ANT / CAL state
payload	N × 16	IQ8 (or packed IQ10)
CRC	16	bit errors on the parallel bus

If one lane slips by a clock, the sync word no longer lines up across the 16 bits and the S31 sees it at once. Bit transposition is cheap on a PC and expensive on the S31 at 40 M words/s.

C5 TRM v1.1 §43: PARLIO ≤ 40 MHz, external TX clock, not an ETM peripheral; a gated external clock may start after TX_START. Lockstep across chips is an inference to test.

Two data modes

Mode 1 — continuous reduced rate

Every C5 streams a narrow band continuously over its own 2-bit lane.

Sample rate (nearest ESP-SDR rate)	4 MS/s, IQ8
Per element	64 of 80 Mbit/s
Per tile / array	512 Mbit/s / 1.02 Gbit/s
Instantaneous bandwidth	≈ 4 MHz
Doppler at 5.2 GHz	34.7 Hz per m/s

- 5 MS/s is not an ESP-SDR rate, and 8 MS/s (128 Mbit/s) does not fit a 2-bit lane.
- The C5 has no continuous capture today. Back-to-back 4 MS/s snapshots (4.1 ms each, 3.3 ms to drain) would be near-continuous if capture and transfer can overlap.
- The C5 analog filter is at least 11 MHz wide: at 4 MS/s out-of-band signals may alias (the S31 stream subsamples without a decimation filter).
- On the S31, ESP-SDR streams by routing the modem debug bus into PARLIO. The C5 lists MODEM_DIAG0–31 signals, but IQ on them is undocumented — and 16–20 lanes per C5 would not fit the shared bus.

Mode 2 — full-rate synchronized snapshots

All 16 elements capture the same 205 µs at 80 MS/s, then drain over their lanes.

Snapshot	16,380 samples, 205 µs
Size per element, IQ8 / IQ10	32.8 / 41.0 kB
Drain time at 80 Mbit/s	3.3 / 4.1 ms
Max snapshot rate (IQ8)	≈ 287 per second
Array data per snapshot	524 kB

- Wide instantaneous bandwidth (11–48 MHz analog) with low-cost networking.
- Right mode for calibration, wideband channel sounding and radar trials.
- Needs PARLIO export firmware and a hardware capture trigger — both new work.

Confirmed Engineering inference Open — needs bench test

ESP-SDR C5 limits: rates 80/40/20/10/8/4 MS/s, ≤ 16,380 samples per snapshot, 11–48 MHz analog bandwidth. Lane = 2 bit × 40 MHz = 80 Mbit/s.

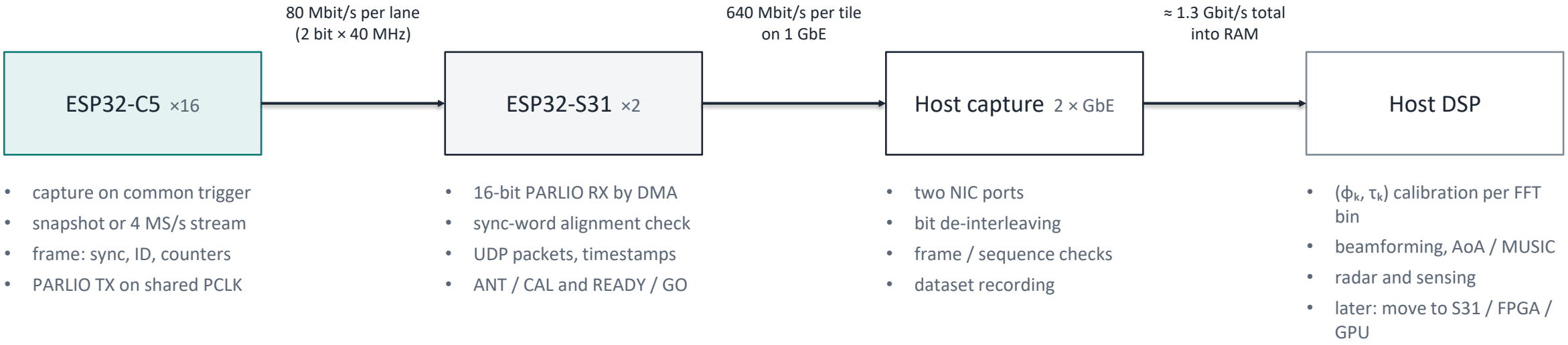
Data-rate budget, stage by stage

Stage	per element	per tile (×8)	array (×16)	comment
Internal IQ, 80 MS/s, IQ8	1.28 Gbit/s	10.2 Gbit/s	20.5 Gbit/s	inside the chip only
Snapshot, 16,380 samples (IQ8)	32 KiB	256 KiB	512 KiB	205 µs of signal
Today: USB-Serial-JTAG	12 Mbit/s	—	—	ESP-SDR, line rate
PARLIO lane, 2 bit × 40 MHz	80 Mbit/s	640 Mbit/s	1.28 Gbit/s	64 % of GbE per tile
Mode 1: 4 MS/s IQ8 continuous	64 Mbit/s	512 Mbit/s	1.02 Gbit/s	C5 continuous capture is open
Mode 2: snapshot drain (IQ8)	3.3 ms	same, parallel	same, parallel	≤ 287 snapshots/s
64 FFT bins per snapshot, 100/s	205 kbit/s	1.64 Mbit/s	3.28 Mbit/s	complex, 16-bit I/Q
16 × 16 covariance, 100/s	—	—	870 kbit/s	136 complex float32

Per tile the lanes add up to 640 Mbit/s and one Gigabit port; the array needs two ports at the host. Reduction (FFT bins, covariance) can move to the S31 or an FPGA later.

Assumptions: IQ8, 16,380-sample snapshots (ESP-SDR maximum on C5), 2 bit × 40 MHz per lane. Replace with step-1/2 measurements.

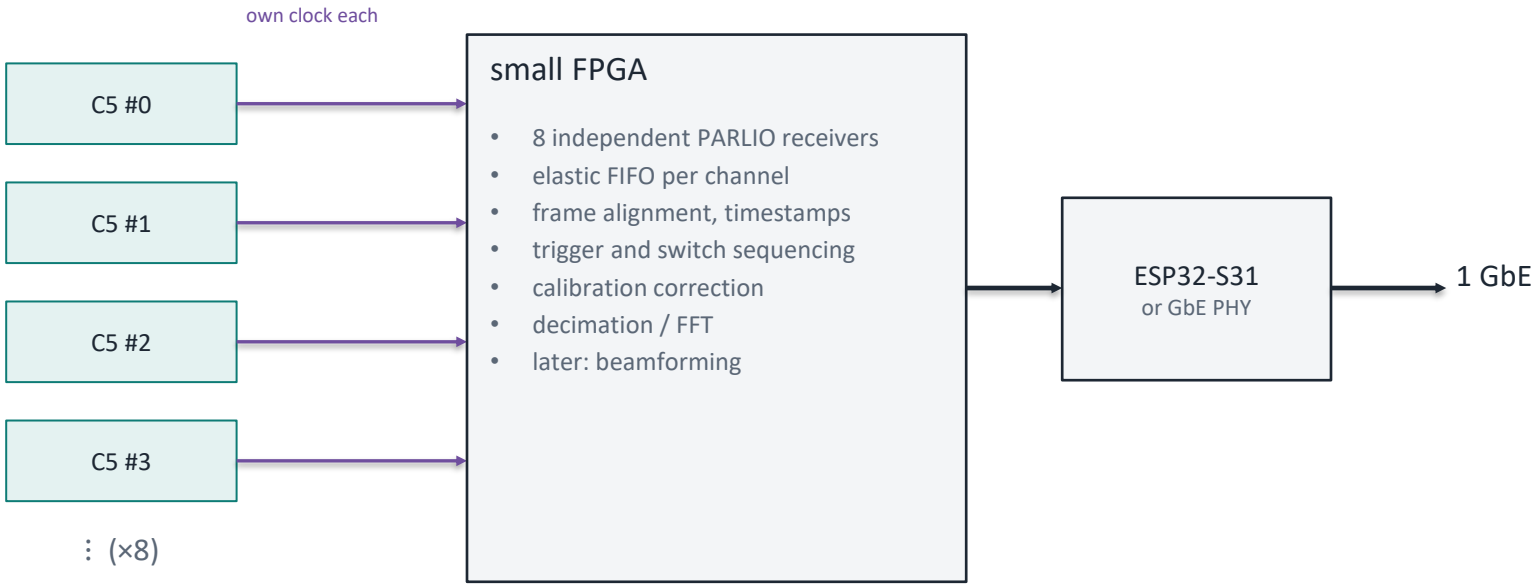
What happens to the IQ data?



The bottleneck is still inside each C5 — capture memory and how fast the IQ reaches PARLIO. Everything after the S31 is ordinary Gigabit networking; the host does the heavy DSP at first.

Lane = 2 bit × 40 MHz. S31 PARLIO→GbE at 640 Mbit/s is shown by ESP-SDR's S31 streaming firmware (one board, internal source).

Plan B: a small FPGA between the C5s and the S31



Switch to Plan B if

- eight C5 lanes cannot be kept in lockstep
- the S31 cannot sustain 640 Mbit/s in and out
- real-time per-element processing (FFT, decimation) is needed

What it costs

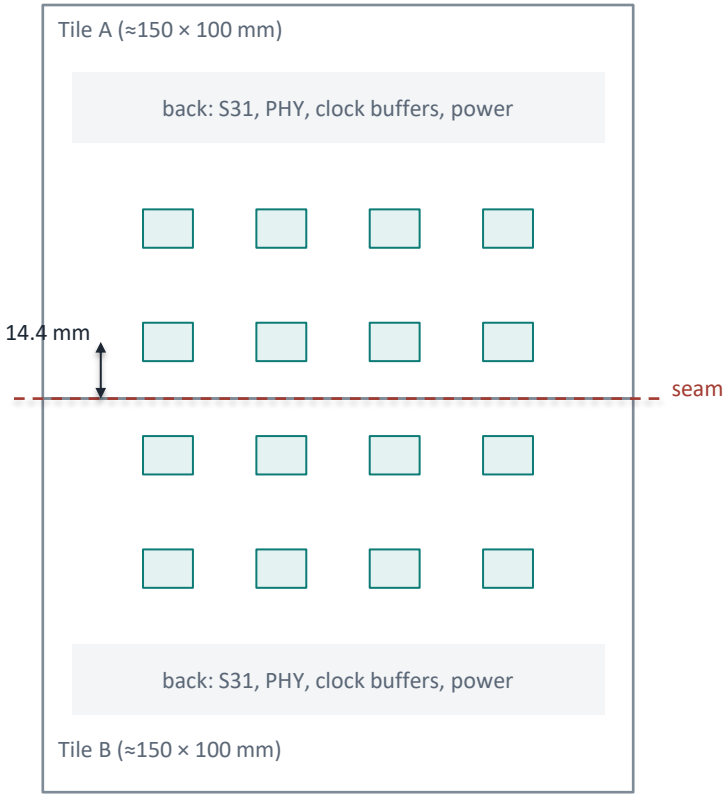
- one more chip, toolchain and HDL work
- more power and board area
- the team needs FPGA experience

What it adds

- each C5 can run on its own lane clock
- deterministic, inspectable timing
- data reduction before Ethernet
- a path to real-time beamforming

Candidate FPGAs: Lattice ECP5 / iCE40 UltraPlus (open toolchain), Efinix Trion. Optional for Rev 1.

Tile PCB: geometry, seam and stack-up



To scale: the 2 × 4 antenna block sits against the seam. Front: patches only. Back: each C5 under its own patch, the rest on the outer half. Rows keep 28.8 mm pitch across the seam only if each board edge is 14.4 mm from its nearest row.

Rev 1 stack-up (custom 4-layer FR-4)

L1	patches only (the front)
L2	solid ground (patch ground)
L3	ground; clock lines, 3.3 V spine
L4	all parts, RF, data lines

Fab cost (team estimate)

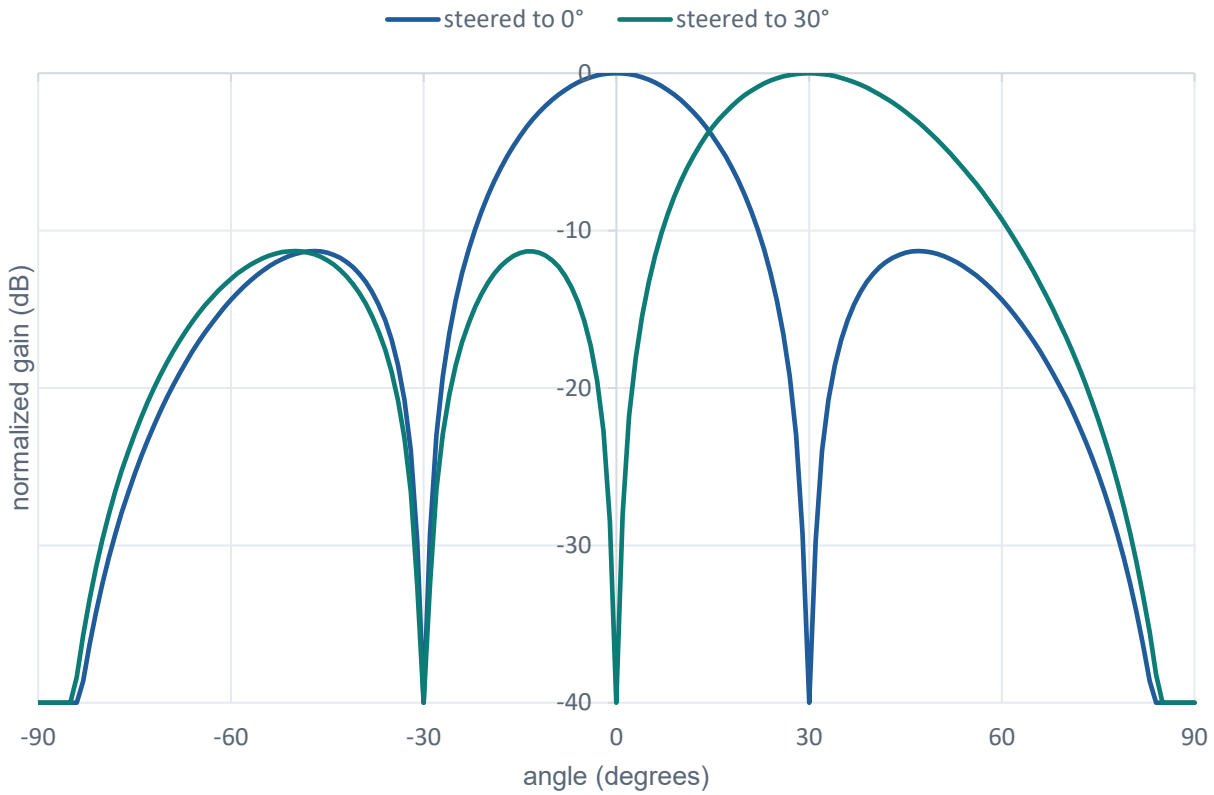
standard 4-layer, 5 boards	$25–35
+ ENIG, impedance control	$35–55
Rev 1 custom stack + blind vias	quote
per tile (standard, 5 boards)	$7–11
6-layer, per board	$10–16
shipping to Israel	$20–35

The patch substrate

- A standard 1.6 mm 4-layer board has only ≈ 0.21 mm between L1 and L2: a 5.2 GHz patch there radiates at most ≈ 18 % of its power on FR-4 (lossless bandwidth ≈ 0.3 %; the loss makes the S11 dip look wider than it is).
- Rev 1 uses a custom stack with a 1.065 mm core under L1: ≈ 1.6 % (≈ 85 MHz) lossless bandwidth and at most ≈ 53 % efficiency; patch ≈ 17.5 × 13.5 mm, trimmed after a test coupon (≈ 55 MHz per 0.1 of εr).
- Front: patches only (plus tented through-via pads). Each C5 sits behind its own patch: probe via, 1 mm to the SPDT, 2.5 mm to the C5, the same for all eight. All digital lines run behind the L2 ground.
- Seam: the ground is cut between tiles, so bridge it continuously and calibrate per element. Tile B is turned 180°, so elements 9–16 carry a fixed 180° term.

To scale for 28.8 mm pitch. Custom 4-layer stack (JLCPCB's standard one has only 0.21 mm under L1). Patch: εr 4.4, tan δ 0.02; lossless BW ≈ $3.77\cdot(\varepsilon_r-1)/\varepsilon_r^2\cdot(W/L)\cdot(h/\lambda_0)$; efficiency counts dielectric loss only. Prices: team estimates.

First useful experiment: receive beamforming



Array factor of one 4-element row at λ/2, uniform weights. Half-power beamwidth ≈ 25° at broadside, first sidelobe −11.3 dB. The full 4×4 array steers in both azimuth and elevation.

Experiment

- Place a 5.2 GHz CW or Wi-Fi source in the far field ($2D^2/\lambda \approx 0.9$ m for the 163 mm aperture diagonal; use ≥ 1 m).
- Calibrate with the internal tone, then capture snapshots with the source at known angles.
- Form beams digitally: $y = w^H x$ with w = steering vector; scan azimuth and elevation.
- Estimate angle with conventional beamforming and MUSIC; add a second source for nulling.

Success criterion

The measured beam pattern matches the predicted one to within a few dB in the main lobe.

The estimated angle repeats after a power cycle and retune — proving that calibration, not luck, produced coherence.

Array factor computed for ideal isotropic elements; real embedded patterns and coupling will differ.

Radar experiments

	Experiment	What we would measure	Evidence and dependencies
●	Wi-Fi sensing from CSI	motion, presence, gestures, breathing-scale phase	widely shown on ESP32; C5 CSI gives 52–242 subcarriers
○	Passive radar / localization with AP illuminators	AoA, Doppler, multipath changes, position	ESPARGOS passive localization at 2.4 GHz; needs mode 1 or fast snapshots
○	Separate TX and RX elements	1–2 C5s transmit packets, 14–15 receive coherently	standard packet TX; ESPARGOS radar mode does one-TX / rest-RX
○	Pulsed / switched (TDM)	transmit a burst, switch to RX, capture echoes	C5 is half-duplex; TX→RX switching time unknown
◌	Active CW or stepped frequency	Doppler (CW), coarse range (stepping)	5 GHz tone from a C5 undocumented
◌	FMCW-like on the same radio	range–Doppler maps	needs simultaneous TX + RX and arbitrary TX — not the starting point

Doppler at 5.2 GHz is 34.7 Hz per m/s; 0.5 m/s resolution (walking people) needs ≈ 60 ms of coherent slow time. Mode 1 at 4 MS/s or back-to-back snapshots make this possible; USB-only snapshots do not.
Range resolution is c/2B: 7.5 m at 20 MHz, ≈ 3.1 m at the C5's 48 MHz analog limit. Angle and Doppler are the strengths.

● Confirmed ○ Engineering inference ◌ Open — needs bench test ◌ Speculative

Status = evidence on ESP32-class hardware. Literature: Chetty 2012, Colone 2011, Yildirim 2020, Euchner 2025 (passive channel charting). $f_D = 2v/\lambda$; $\Delta r = c/2B$.

Communications experiments

	Experiment	Content	Builds on
○	Receive beamforming	steer a digital beam to a Wi-Fi station; measure SNR gain against one element	calibrated snapshots
○	Interference nulling	two transmitters; place a null on one, keep the other	covariance estimation
○	Wi-Fi angle of arrival	AoA of real stations from per-packet CSI — shown at 2.4 GHz by ESPARGOS	CSI mode + calibration
○	MIMO channel measurements	16-element channel matrices, multipath, angle–delay profiles	CSI or raw snapshots
○	Localization and beam tracking	follow a moving station; compare with ground truth	repeated snapshots / CSI
◌	Distributed coherent TX	beam-steered transmission from several C5s	TX phase control (open)

● Confirmed ○ Engineering inference ◌ Open — needs bench test ◌ Speculative

Receive-side items need calibrated coherent capture or CSI; the TX item depends on the open questions on the next slide.

Transmit: what we know and what we don't

	Capability	Evidence / comment
●	Standard Wi-Fi packet TX at 5 GHz on every C5	datasheet: 18 dBm typ. at low rates (chip level)
●	One element transmits, the others receive (TDM)	ESPARGOS One radar mode does this at 2.4 GHz
○	All TX carriers on the same frequency	follows from the shared 48 MHz reference
◌	Known, repeatable TX LO phase after each PLL lock	unknown; RX calibration does not measure the TX path
◌	Simultaneous TX start on 16 chips	packet TX timing belongs to the Wi-Fi MAC, not to us
◌	CW tone at 5 GHz on the C5	cert-test API esp_phy_wifi_tx_tone() is documented for channels 1–14 only
◌	Arbitrary IQ playback on the C5	ESP-SDR is receive-only; its authors say arbitrary TX works but are not releasing it (misuse and certification concerns)
◌	16-element coherent transmit beamforming	needs all of the above plus over-the-air TX calibration

Regulation: 5150–5250 MHz allows 200 mW EIRP, indoor, in both the EU and Israel, for EN 301 893 RLAN equipment. Sixteen coherent elements add up to 24 dB over one, and custom radar waveforms are probably not covered — do TX experiments cabled, shielded or at low power, on channels 36–48.

● Confirmed ○ Engineering inference ◌ Open — needs bench test ◌ Speculative

C5 datasheet v1.5; ESP-SDR README, issue #3; ESP-IDF esp_phy_cert_test.h; ESPARGOS One manual §9.4; ECC/DEC/(04)08; Israeli regs 2021.

Transmit steering: easy maths, hard hardware

Phase step between neighbours at λ/2

$$\Delta\varphi = -\pi \cdot \sin\theta, \quad s_k[n] = s[n] \cdot e^{jk\Delta\varphi}$$

steering angle θ	Δϕ per element
0°	0°
15°	−46.6°
30°	−90.0°
45°	−127.3°
60°	−155.9°

What is actually hard

- Phase-coherent TX LOs: a shared 48 MHz fixes frequency, not the phase after each PLL lock.
- Deterministic TX start: normal packet TX is timed by the Wi-Fi MAC, not by us.
- A raw TX waveform path on the C5 — the ESP-SDR authors say arbitrary TX works but have not released it.
- The C5 is 1T1R and half-duplex: one chain switches between TX and RX.

If steering is needed before that is solved

- Separate roles: 1–2 C5s transmit, the other 14–15 receive coherently.
- TDM / MIMO: one element transmits at a time, the TX aperture is synthesized afterwards (as in ESPARGOS radar mode).
- External RF phase shifters after each C5 — more BOM, loss and control lines; not for Rev 1.

Phase steps for λ/2 spacing. ESPARGOS states phase-coherent TX works on ESPARGOS One but has not released it (ESP-SDR page, issue #3).

Power, heat and board constraints

Supply estimate (3.3 V rail)

Load	per unit	count	total
ESP32-C5, receive (5 GHz)	110 mA	16	1.76 A
ESP32-C5, transmit peak (5 GHz)	408 mA	16	6.53 A
ESP32-S31 + RTL8211F (estimate)	≈ 400 mA	2	0.80 A
Clock buffers, oscillators, SPDTs	≈ 40 mA	6	0.24 A
Master board incl. synthesizer	≈ 200 mA	1	0.20 A
All receiving			3.0 A · 10 W
All transmitting (worst case)			7.8 A · 26 W

- Espressif asks for ≥ 0.6 A of supply capability per C5: 4.8 A per tile. Plus 10 µF on each VDDA rail.
- TX bursts droop a shared rail and can disturb the reference — give clock buffers and oscillators their own quiet LDO.
- Heat: ≈ 10 W in receive over two tiles is easy; continuous all-TX (26 W) needs airflow.

Board-level points

- Two tile PCBs, custom 4-layer FR-4 with controlled impedance; one master timing / RF board.
- Front: patches only. Each chip-down ESP32-C5 sits behind its own patch, so XTAL_P can take the shared 48 MHz.
- PARLIO lines on L4, series-terminated; ≈ 1.2 ns of layout skew (data plus PCLK) and the C5 clock-to-out spread (a bench test) against the 12.5 ns half-period.
- 48 and 40 MHz point-to-point from 1:8 / 1:10 buffers: on L3 from the buffers, the last few cm to each C5 on L4; series-terminated, away from RF lines.
- RF: 50 Ω lines on L4 over a solid L3 ground; each SPDT 1 mm from its probe via and 2.5 mm from its C5.
- Antenna substrate: the 1.065 mm core under L1 (see the tile PCB slide).

C5: datasheet v1.5 Table 5-8 (peak, 3.3 V). S31, PHY and master-board figures are planning estimates — measure in steps 2 and 4.

What a tile and the full array cost

One 8-element tile (electronics)

Item	≈ USD
8 × ESP32-C5 (chip-down needed for a shared reference)	40
1 × ESP32-S31	7.5
1 × RTL8211F Gigabit PHY	1.4
Gigabit MagJack	2.5
48 / 40 MHz fan-out buffers	6
40 MHz oscillator	1.5
8 × RF SPDT (SKY13351-class)	5
3.3 V regulation, power	3
Ethernet crystal, passives, misc.	3–5
Tile electronics	70–80

Full 16-element system

Item	≈ USD
Two tiles	140–160
Master board: 48 MHz, trigger, 5.2 GHz source, splitters	15–20
Electronics total	160–180
PCBs, assembly, shipping	60–120
Complete prototype target	220–300

- Chip-down C5s replace modules because modules carry their own crystal; parts cost is similar, layout work is higher.
- The custom stack-up (with blind L4–L3 vias) keeps the patches on the tile but needs a quote: the PCB line assumes standard-stack prices. Ask for a balanced build and a bow/twist limit. A separate antenna board would add a second fab order and 16 RF connections.
- A MAX2871 evaluation board alone costs more than the $15–20 line; a bare synthesizer on the master board is assumed.
- For scale: an ESPARGOS One Core Kit (8 elements, 2.4 GHz, finished product with enclosure, PoE and firmware) is €350 excl. VAT.

Team low-volume estimates (Oct 2026), not quotes. Excludes lab instruments, enclosure and engineering time.

Software stack

ESP32-C5 firmware	ESP32-S31 firmware	Host	Later
• ESP-SDR capture (pinned IDF) • fixed gain, RF tuning • trigger input, READY • pack IQ8 / IQ10 • PARLIO TX by GDMA • frame header: sync, ID, counters	• 16-bit PARLIO RX by DMA • lane alignment check • READY / GO, ANT / CAL • UDP packetization • PTP timestamps • control API	• two GbE capture threads • bit de-interleaving (NumPy) • frame / sequence checks • calibration (ϕ_k, τ_k per bin) • beamforming, MUSIC, radar • Python, SciPy; MATLAB optional	• reduction on the S31 • FPGA preprocessing (Plan B) • GPU (CuPy) for real-time • GNU Radio blocks • dataset publishing

Visualization

beam pattern	AoA / MUSIC spectrum	spectrogram	range–Doppler map	per-element phase vs time	calibration health

Every captured snapshot is stored with its calibration state, so results can be re-processed offline.

Everything above the C5 firmware is ordinary open tooling; the C5 layer depends on ESP-SDR and ESP-IDF internals.

Critical bench experiments before the tile PCB

	Test	What it proves	Evidence today
◌	C5 raw IQ → GDMA → PARLIO TX	a snapshot can leave the C5 without CPU copies	GDMA can address the ESP-SDR IQ bank; ESP-SDR packs IQ8 in place on the CPU; access while the modem owns the bank is unknown
◌	2 bit × 40 MHz from one C5	80 Mbit/s per element in practice	40 MHz is PARLIO's maximum clock; receiver samples on the falling edge (≈ 12.5 ns window)
◌	eight C5 PARLIO transmitters in lockstep	Rev 1 needs no FPGA	PARLIO has no ETM start; the reference manual's gated external clock suggests a clock-held start — untested
○	16-bit PARLIO RX at 40 MHz into the S31	tile ingress	ESP-SDR S31: 16-lane PARLIO → GDMA ring at 40 MS/s, but fed internally with an internal sample clock — external pins + PCLK untested
●	S31 → GbE at ≈ 640 Mbit/s	tile egress	ESP-SDR: raw UDP frames at ≈ 687 Mbit/s, 600 s, 0 samples lost (one board, experimental)
◌	shared 48 MHz into C5 XTAL_P	frequency lock across elements	no C5 guidance; ESP32-S2 oscillator note, ESPARGOS at 40 MHz
◌	RF phase repeatability: reset, retune, temperature	how often to calibrate	ESPARGOS: phase re-rolls at 2.4 GHz; C5 unmeasured
◌	snapshot synchronization on a wideband signal	real phase, delay and jitter	ESPARGOS One: ≤ 37.5 ns timestamp scatter (2.4 GHz, closed firmware)

● Confirmed ○ Engineering inference ◌ Open — needs bench test

Status = what is already stated or shown in a primary source, not on our hardware. Evidence: ESP-SDR source and docs, ESP32-C5 / S31 datasheets, ESPARGOS papers.

Development sequence

1 One C5

- run ESP-SDR
- characterise raw IQ
- inspect memory / DMA path
- implement PARLIO export

Done when

a snapshot leaves over PARLIO, bit-exact

2 C5 + S31

- 2-bit PARLIO, shared 40 MHz
- ≈ 80 Mbit/s stream
- out over GbE

Done when

80 Mbit/s to the PC for hours, no loss

3 Two C5s

- common 48 MHz reference
- common trigger
- phase calibration
- first AoA

Done when

relative phase stable to a few degrees

4 Eight C5s

- full tile
- 16-bit PARLIO, ≈ 640 Mbit/s
- receive beamforming

Done when

8 lanes aligned for hours; beam matches prediction

5 Second tile

- 16 elements
- common calibration
- 4 × 4 AoA and beamforming

Done when

calibrated 2D scan, repeatable after power cycle

6 Extensions

- radar, Wi-Fi sensing
- coherent TX research
- digital TX beamforming
- FPGA preprocessing

Done when

research results

Each step ends with a measurable result; nothing larger is built until the previous step passes.

Main technical risks

Risk	Early test	Fallback
C5 raw IQ cannot reach PARLIO by DMA at the needed rate	step 1: snapshot → PARLIO, bit-exact	CPU copy at lower rate; snapshots only
Eight lanes not cycle-aligned (40 MHz is PARLIO's limit)	steps 2 and 4: sync-word checks, eye margin	Plan B: FPGA with per-lane FIFOs
S31 cannot ingest and send 640 Mbit/s at once	step 2 → 4: 80 → 640 Mbit/s	lower lane rate, or FPGA → GbE PHY
Shared 48 MHz gives poor phase stability	step 3: phase vs time and temperature	per-snapshot calibration
C5 LO phase not repeatable across resets / retunes	step 3: repeated resets and retunes	calibrate every epoch (as ESPARGOS)
Coherent TX method not reproducible on C5	step 6, research only	RX-only results stand alone
FR-4 antenna / array consistency insufficient	VNA on the first tile	separate aperture board (ours) or better laminate
Power and heat with 16 active C5s	current and temperature on one tile	duty-cycle TX, airflow
No continuous capture on the C5 (mode 1)	step 1: back-to-back snapshot timing	mode 2 only; quasi-continuous snapshots
Custom stack: cost, warpage; patches lossy (≤ 53 %)	quote; simulate + VNA one tile	standard stack + our aperture board

The first eight are the team's list; the last two came out of the research. Each except coherent TX (step 6) has a bench test in steps 1–4.

What we know, what we infer, what we must test

	Claim	Basis
●	ESP32-C5 receives 5180–5885 MHz; 1T1R; 18 dBm TX, −94 dBm sensitivity (low rates)	Espressif datasheet v1.5
●	ESP-SDR raw IQ on C5: 80/40/20/10/8/4 MS/s, ≤ 16,380 samples, USB / UART out	ESP-SDR source and README
●	C5 PARLIO TX: 1/2/4/8 bit, ≤ 40 MHz, external clock pad, GDMA; not an ETM peripheral	C5 TRM v1.1, ESP-IDF source
●	S31 streams 16-lane PARLIO capture to GbE at ≈ 687 Mbit/s (experimental)	ESP-SDR S31 docs; 600 s, 0 lost
●	S31 EMAC: 10/100/1000, MII / RMII / RGMII, IEEE 1588 timestamps	S31 datasheet v0.5 (pre-release)
●	ESP32-C5 modules integrate their own 48 MHz crystal	module datasheet v1.3
●	ESP32 PLL phase changes after reset and retune	ESPARGOS hardware paper
○	2-bit lanes from eight C5s can be packed into one 16-bit S31 bus	PARLIO widths and clocks
○	A 2-bit lane carries 4 MS/s IQ8 continuously (64 of 80 Mbit/s)	arithmetic
◌	C5 IQ buffer → GDMA → PARLIO without CPU copies	untested
◌	Eight C5 PARLIO transmitters stay in lockstep	untested
◌	XTAL_P can be driven cleanly from a shared 48 MHz buffer	not documented for C5
◌	Continuous IQ capture on the C5	no C5 continuous mode today
◌	Coherent TX on C5	ESPARGOS: works on ESPARGOS One, unpublished
◌	16-element coherent TX beamforming	no evidence yet

● Confirmed ○ Engineering inference ◌ Open — needs bench test ◌ Speculative

Checked 12–13 Oct 2026 against the references; open items become bench tests in phases 1–2.

Alternatives we considered

Alternative	Why not for Rev 1	Reconsider when
W5500 SPI-Ethernet per C5	C5 has no MAC; adds 8 bridges, PHYs, magnetics and a switch; 100 Mbit ceiling per element	single-node experiments, elements on separate boards
Ethernet switch per tile (BCM53202M: 8 × 10/100 + 2 × GbE)	needs Ethernet on every C5 first; solves aggregation, not sample synchronization	elements become separate networked boards
ESP32-S31 as the RF element (2.4 GHz)	continuous IQ works today, but 2.4 GHz only: new aperture, crowded band	if C5 PARLIO export fails
4 × 2×2 tiles, one FPGA each	more boards, FPGA mandatory, four uplinks	Plan B, or more processing per tile
Analog phase shifter per element	receive steering is exact and wideband in software	deterministic TX steering before raw TX exists
Buy 2 × ESPARGOS One (2.4 GHz)	closed IQ firmware, 2.4 GHz, not our design	as a reference instrument and baseline

What a successful student project looks like

Minimum

a solid result even if the tile PCB is late

- C5 raw IQ exported over PARLIO, measured
- C5 + S31 streaming to a PC over GbE
- two C5s on one reference, phase-calibrated
- open-source code + measurement report

Target

the proposal as planned

- one 8-element tile at 640 Mbit/s
- two tiles: calibrated 4 × 4 beamforming and AoA
- both data modes working
- a published dataset of synchronized captures

Stretch

only after the target is met

- Doppler / passive radar demo
- separate-TX bistatic experiments
- first coherent-TX experiments
- FPGA or S31 preprocessing

A good outcome is measured coherence and a reusable platform — not a single beam plot. Every result should survive a power cycle.

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